

PLUS-PURE THRESHOLDS OF SOME CUSP-LIKE SINGULARITIES IN MIXED CHARACTERISTIC

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ABSTRACT. Log-canonical and F -pure thresholds of pairs in equal characteristic admit an analog in the recent theory of singularities in mixed characteristic, which is known as the plus-pure threshold. In this paper we study plus-pure thresholds for singularities of the form $p^a + x^b \in \mathbf{Z}_p[[x]]$, showing that in a number of cases this plus-pure threshold agrees with the F -pure threshold of the singularity $t^a + x^b \in \mathbf{F}_p[[t, x]]$. We also discuss a few other sporadic examples.

1. INTRODUCTION

Let X be a smooth algebraic variety over $\mathbf{C}$, let $x \in X$ be a closed point and $D = (f = 0)$ be a prime divisor containing x . If $R := \widehat{\mathcal{O}_{X,x}}$ denotes the completion of the local ring at x , the log-canonical threshold $\text{lct}(R, f)$ is the threshold in t at which the pair (X, tD) becomes not log canonical at x ; in other words, it is the first jumping number of the multiplier ideal $\mathcal{J}(R, f^t)$. This invariant has been used prominently in birational algebraic geometry and the minimal model program; see [Kol97, Laz04] for some applications.

Log canonical pairs are closely related to F -pure pairs in characteristic $p > 0$ [HW02, MS11, BST13]. Inspired by this connection, Takagi and Watanabe defined the F -pure threshold $\text{fpt}(R, f)$ as the threshold in t at which (R, f^t) becomes not F -pure [TW04], cf. [MTW05, HMTW08]. When R is regular, this is also the first jumping number of the test ideal $\tau(R, f^t)$, a characteristic- p analog of the multiplier ideal. Note that there has been a great deal of effort directed towards computing the F -pure thresholds of specific (classes of) polynomials [ST09, Her15, Her14, BS15, MSV14, HNNBWZ16, HT17, DSNnB18, Mü18, Pag18, KKP⁺22, SV23].

In recent years, building upon the theory of Scholze's perfectoid algebras [Sch12] and the work of André, Bhatt, Gabber, Heitmann-Ma and others [And18, Bha18, And20, Gab18, HM18, Bha20], the development of a theory of singularities in mixed characteristic is now underway [MS18, MS21, DT21, MST⁺22, BMP⁺23a, Rob22, HLS24, Mur21, CLM⁺22, BMP⁺23b, BMP⁺24, RV24]. This theory provides an analog for the multiplier and test ideals. The first jumping number—known as the *plus-pure threshold*—therefore provides an analog of the log-canonical and F -pure thresholds. A succinct definition is as follows: if $(R, \mathfrak{m})$ is a regular complete local domain of mixed characteristic, $0 \neq f \in R$ is an element, and $t \in \mathbf{Q}_{>0}$ is a positive rational number, we denote by $f^t : R \rightarrow R^+$ the unique R -linear map sending $1 \in R$ to a choice of $f^t \in R^+$; then the plus-pure threshold of (R, f) is given by

$$\text{ppt}(R, f) := \sup\{t \in \mathbf{Q}_{>0} \mid f^t : R \rightarrow R^+ \text{ splits}\},$$

where R^+ is the absolute integral closure of R (see Lemma 2.2 for other characterizations).

Except for some toric-type singularities, see for instance [Rob22], essentially no examples of plus-pure thresholds have been calculated (although some upper and lower bounds were known, and sometimes they coincide, see below). This paper begins this exploration by computing the plus-pure thresholds of some cusp-like singularities $f := p^a + x^b \in \mathbf{Z}_p[[x]]$, where $\mathbf{Z}_p$ is the ring of p -adic integers. Indeed, unlike characteristic $p > 0$, one cannot simply use Frobenius and so the computations become much more subtle.

Before we discuss our work, we remark that something was known about the examples $f := p^2 + x^3 \in \mathbf{Z}_p[[x]]$ and $f := p^3 + x^2 \in \mathbf{Z}_p[[x]]$. For either choice, by [BMP⁺23a, Proposition 4.17, Lemma 4.18] or [MS21, Theorem 6.21] and [Bha20], if the map $\mathbf{Z}_p[[x]] \rightarrow \mathbf{Z}_p[[x]]^+$ given by $1 \mapsto f^t$ were to split for some $t \in \mathbf{Q}_{>0}$, the pair $(\mathbf{Z}_p[[x]], f^t)$ would be log-terminal, then a blowup computation shows that $t < 5/6$ and therefore $\text{ppt}(\mathbf{Z}_p[[x]], f) \leq 5/6$. On the other hand, the F -pure threshold of the related polynomial $f_0 := y^2 + z^3 \in$

$\mathbf{F}_p[[y, z]]$ provides a lower bound for $\text{ppt}(\mathbf{Z}_p[[x]], f)$; indeed, combining [MST⁺22, Example 7.10] with [Bha20], we see that $f^t : \mathbf{Z}_p[[x]] \rightarrow \mathbf{Z}_p[[x]]^+$ splits whenever $(\mathbf{F}_p[[y, z]], f_0^t)$ is F -regular. We conclude that

$$\text{fpt}(\mathbf{F}_p[[y, z]], f_0) \leq \text{ppt}(\mathbf{Z}_p[[x]], f) \leq \frac{5}{6},$$

and we recall that the lower bound $\text{fpt}(\mathbf{F}_p[[y, z]], f_0)$ is known to be as follows (see [MTW05, Example 4.3]):

$$\text{fpt}(\mathbf{F}_p[[y, z]], f_0) = \begin{cases} 1/2 & \text{if } p = 2 \\ 2/3 & \text{if } p = 3 \\ (5p-1)/6p & \text{if } p \equiv 5 \pmod{6} \\ 5/6 & \text{if } p \equiv 1 \pmod{6}. \end{cases}$$

In particular, it was known that $\text{ppt}(\mathbf{Z}_p[[x]], f) = 5/6$ when $p \equiv 1 \pmod{6}$.

As our first contribution, we show that analogous upper and lower bounds hold for any cusp-like polynomial $f = p^a + x^b \in \mathbf{Z}_p[[x]]$; this is verified in detail in Proposition 2.9 by utilizing some of the main results of [MS21, MST⁺22, BMP⁺23a].

Our main goal is then to show that, in many cases, one has $\text{ppt}(R, f) = \text{fpt}(R_0, f_0)$; in other words, that one has equality on the lower bound. For example, we prove that this is the case for the polynomials $f = p^2 + x^3$ and $f = p^3 + x^2$ discussed above. Note that, by Proposition 2.9 as mentioned above, one has $\text{ppt}(f) = \frac{1}{a} + \frac{1}{b}$ whenever $\text{fpt}(R_0, f_0) = \frac{1}{a} + \frac{1}{b}$, and some of the remaining cases are handled by the theorem below.

Main Theorem (Corollary 3.2). *Fix a prime $p > 0$, let $R := \mathbf{Z}_p[[x]]$ be a power series ring in one variable over the p -adic integers, and let $R_0 := \mathbf{F}_p[[y, x]]$ be a power series ring in two variables over $\mathbf{F}_p$. Fix integers $a, b \geq 2$ and consider the polynomials $f := p^a + x^b \in R$ and $f_0 := y^a + x^b \in R_0$.*

Assume that $\text{fpt}(R_0, f_0) \neq \frac{1}{a} + \frac{1}{b}$, and fix an integer $e \geq 1$ such that $p^e \text{fpt}(R_0, f_0) \in \mathbf{Z}$. If we have

$$\text{fpt}(R_0, f_0) \geq \frac{1}{a} + \frac{1}{b} - \frac{1}{ap^e},$$

then $\text{ppt}(R, f) = \text{fpt}(R_0, f_0)$.

Note that an algorithm for the computation of $\text{fpt}(R_0, f_0)$ is provided in [Her14, Her15], cf. [ST09] (we briefly discuss this algorithm in Section 3). It follows from some of the references above that, unless one has $\text{fpt}(R_0, f_0) = \frac{1}{a} + \frac{1}{b}$, the denominator of $\text{fpt}(R_0, f_0)$ is a power of p , and therefore one can always choose an $e \geq 1$ as in the statement of the theorem.

Since the F -pure threshold $\text{fpt}(R_0, f_0)$ can be computed following the algorithm mentioned above, our result gives an effective method for finding values of p, a, b for which we have $\text{ppt}(p^a + x^b) = \text{fpt}(y^a + x^b)$. Using the slightly stronger version of the Main Theorem given in Corollary 3.2, as well as the discussion thereafter, we know that $\text{ppt}(R, p^a + x^b) = \text{fpt}(R_0, y^a + x^b)$ for the primes given in Table 1 (we also prove some more cases in Section 4).

TABLE 1. Values of p for which we know $\text{ppt}(p^a + x^b) = \text{fpt}(y^a + x^b)$.

$a \backslash b$	2	3	4	5
2	All p	All p	All p	All p
3	All $p \neq 2$	All $p \neq 3$	All $p \neq 3$ with $p \not\equiv 11 \pmod{12}$	All $p \neq 3, 5$ with $p \not\equiv 14 \pmod{15}$
4	All p	All $p \neq 3$ with $p \not\equiv 11 \pmod{12}$	All $p \neq 2$ with $p \not\equiv 3 \pmod{4}$	All $p \neq 2$ with $p \not\equiv 3, 19 \pmod{20}$
5	All $p \neq 2, 5$ with $p \not\equiv 7, 9 \pmod{10}$	All $p \neq 3, 5$ with $p \not\equiv 8, 14 \pmod{15}$	All $p \neq 2$ with $p \not\equiv 3, 19 \pmod{20}$	All $p \neq 5$ with $p \not\equiv 2, 4 \pmod{5}$

One can also fix the residue characteristic p , and ask for which values of a, b our main result applies. For $p = 3$, we display these in a scatter plot in Figure 1. While the condition is not dense, one can see there are numerous values of (a, b) where it holds. The plots for other primes look similar.

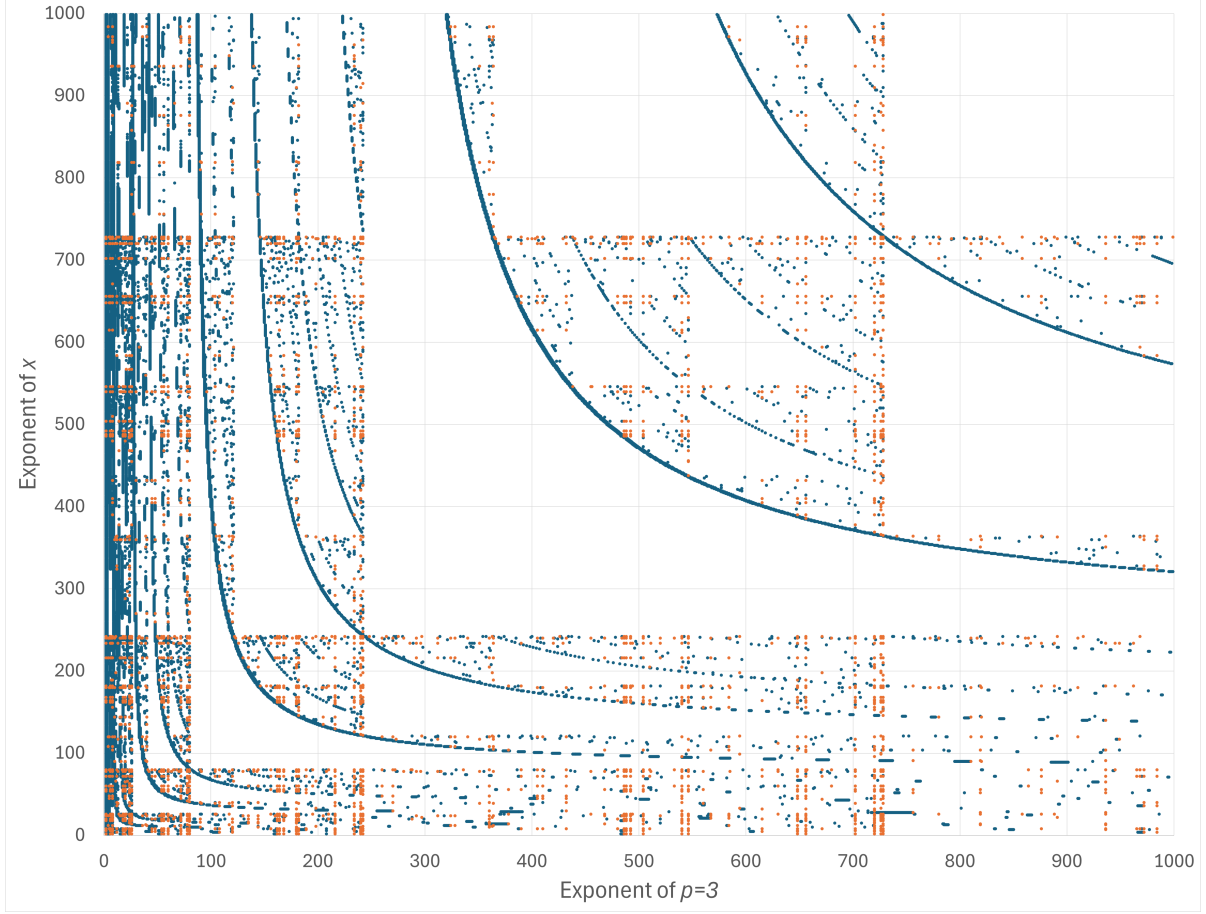


FIGURE 1. Values of (a, b) where we know $\text{ppt}(p^a + x^b)$ when $p = 3$. The orange dots correspond to the a and b values where $\text{fpt}(f_0) = \frac{1}{a} + \frac{1}{b}$. The remaining dots are consequences of Corollary 3.2.

The code to generate this data set is included as an ancillary file to the arXiv version of this document and is also available on Schwede's website.

<https://www.math.utah.edu/~schwede/M2/MakeTable-PPT.m2>

The code is written in Macaulay2 [GS] and uses the FrobeniusThresholds package [HSTW21].

In Section 4, we study more sporadic examples not covered in our main theorem. First, we show that if $f = x^2 + y^3 \in \mathbf{Z}_2[[x, y]] = R$, then $\text{ppt}(R, f) > 1/2 = \text{fpt}(x^2 + y^3 \in \mathbf{F}_2[[x, y]])$, see Corollary 4.3. In Proposition 4.5, we study the ppt of the analog of “three lines”, $px(x - p) \in \mathbf{Z}_p[[x]]$, showing it agrees with the fpt of the analogous three lines. Finally, in Proposition 4.6 we show that the plus-pure threshold of $2^a + x^2 \in \mathbf{Z}_2[[x]]$ is $1/2$ for all integers $a \geq 2$.

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2. PRELIMINARIES

As usual, a prime number $p > 0$ is fixed.

2.1. Plus pure thresholds. Recall a commutative integral domain S is said to be *absolutely integrally closed* if every monic polynomial in $S[T]$ splits into a product of linear factors or, equivalently, if every monic polynomial with coefficients in S has a solution in S . Given an arbitrary integral domain R with fraction field K , after fixing an algebraic closure $\overline{K}$ of K , the integral closure R^+ of R in $\overline{K}$ is an absolutely integrally closed domain, called the *absolute integral closure* of R .

If S is an absolutely integrally closed domain and $z \in S$ is an element, given a nonnegative rational number $\alpha = n/m$ with $n, m \in \mathbf{Z}$, we let $z^\alpha \in S$ denote a choice of element for which $(z^\alpha)^m = z^n$. Note that up to units such a choice of z^α is unique and independent of the representation $\alpha = n/m$. We will be sloppy about the nonuniqueness: in general we will only make statements about the membership of z^α in some ideal of R^+ or statements about ideals in which z^α is a generator. Such statements are, of course, independent of the choice of z^α .

Definition 2.1. Suppose $(S, \mathfrak{m})$ is a complete Noetherian local domain of residue characteristic $p > 0$. We suppose further that $0 \neq f \in \mathfrak{m}$. In this case we define the *plus-pure-threshold* of f to be

$$\text{ppt}(f) := \sup \{ \alpha \in \mathbf{Q}_{\geq 0} \mid S \xrightarrow{1 \mapsto f^\alpha} S^+ \text{ is pure} \}.$$

In this paper, we only study this condition in the case where S is regular.

We write down some equivalent ways to check the purity of the map $S \xrightarrow{1 \mapsto f^\alpha} S^+$.

Lemma 2.2. *With notation as in [Definition 2.1](#) suppose $\alpha \in \mathbf{Q}_{\geq 0}$, and $f^\alpha \in S^+$ is fixed. Then the following is equivalent.*

- (i) *The map $S \xrightarrow{1 \mapsto f^\alpha} S^+$ is pure (equivalently splits).*
- (ii) *If E is an injective hull of $k := S/\mathfrak{m}$, then the induced map $S \otimes_S E \xrightarrow{1 \mapsto f^\alpha} S^+ \otimes_S E$ is injective.*
- (iii) *The map $S \xrightarrow{1 \mapsto f^\alpha} \widehat{S^+}$ is pure (equivalently splits) where $\widehat{S^+}$ is the p -adic (or $\mathfrak{m}$ -adic) completion of S^+ .*
- (iv) *For each finite extension $S \subseteq S' \subseteq S^+$ containing f^α we have that*

$$S \xrightarrow{1 \mapsto f^\alpha} S'$$

splits.

- (v) $\tau_+(S, f^\alpha) = S$ where $\tau_+(S, f^\alpha) = \tau_+(S, \alpha \text{div}(f))$ is the plus-test-ideal denoted as $\tau_{\widehat{S^+}}(S, f^\alpha)$ in [\[MS21\]](#), see also [\[BMP⁺23a, Definition 4.16\]](#), [\[HLS24, Section 4\]](#) (setting $X = \text{Spec } S$), or [\[BMP⁺23b, Section 8\]](#).

If additionally we suppose that S is regular, then these conditions are equivalent to:

- (vi) $f^\alpha \notin \mathfrak{m}S^+$.
- (vii) $f^\alpha \notin \mathfrak{m}\widehat{S^+}$ where $\widehat{S^+}$ is the p -adic (or $\mathfrak{m}$ -adic) completion of S^+

Proof. In parts (i) and (iii) we see that the “equivalently splits” statements are harmless as S is complete so that pure and split maps from S are equivalent, see [\[Fed83, Lemma 1.2\]](#). The equivalence of (i) and (ii)

follows easily from Matlis duality, see [HH95, Lemma (2.1)(e)]. Then the equivalence of (i) and (iii) follows from the fact that $\widehat{S^+} \otimes_S E \simeq S^+ \otimes_S E$ so we win by using (ii). Certainly (iv) and (i) are equivalent since a filtered colimit of split maps is pure and a pure finite map of Noetherian rings is split, see [Sta, Tag 058H].

Unraveling the definition of $\tau_+(S, f^\alpha)$ easily implies that (v) is equivalent to many of the other definitions above. More precisely, since $\tau_+(S, f^\alpha)$ is the Matlis dual of the image of $S \otimes_S E \rightarrow S^+ \otimes_S E$, $\tau_+(S, f^\alpha) = S$ if and only if $S \otimes_S E \rightarrow S^+ \otimes_S E$ is injective.

Clearly, (vi) and (vii) are equivalent since $p \in \mathfrak{m}$, see [Sta, Tag 05GG]. Therefore, to complete the proof, it suffices to prove the equivalence of (vii) and (ii). If $f^\alpha \in \mathfrak{m}\widehat{S^+}$, then $S \rightarrow \widehat{S^+}$ is not pure by tensoring with k . It is for the converse that we finally need regularity. Assume that $f^\alpha \notin \mathfrak{m}\widehat{S^+}$. Since S is regular, $\widehat{S^+}$ is big Cohen-Macaulay and so is faithfully flat over S by [HH92, Page 77] or [Bha20, Lemma 2.9] (it is worth noting that if $S \rightarrow \widehat{S^+}$ is faithfully flat, then S is regular by [BIM19]). Hence the injectivity of $S/\mathcal{M} =: k \rightarrow E$ extends to the injectivity of $k \otimes_S \widehat{S^+} \rightarrow E \otimes_S \widehat{S^+}$, giving the injectivity of $k \rightarrow E \otimes_S \widehat{S^+}$. By considering the socle, we have that $E \rightarrow E \otimes_S \widehat{S^+}$ is injective as desired. $\square$

Remark 2.3. In a previous version of this article, we denoted $\text{ppt}(f)$ by $+\text{pt}(f)$. Because including the $+$ in the notation caused confusion, especially when writing on a board, we have switched to ppt .

The closely related *perfectoid pure threshold of p* , unfortunately also denoted by $\text{ppt}(R, p)$, was defined as

$$\sup \{a/p^e \in \mathbf{Q}_{\geq 0} \mid S \xrightarrow{1 \mapsto \varpi^a} B \text{ is pure for some perfectoid } R\text{-algebra } B \text{ with } \varpi \text{ a } p^e\text{-th root of } p\},$$

see [Yos25].

As that definition can clearly be generalized to any element of S , there is potential for confusion. However, we believe that two notions are the same at least in the context of this paper where $S = W(k)[[x_2, \dots, x_d]]$. The key ingredient is a slight modification of [CLM⁺22, Lemma 5.1.6] which lets one pass from S^+ to perfectoid algebras that can be used to test the perfectoid threshold up to perturbation by f^ϵ for $1 \gg \epsilon > 0$ which will not impact the supremum in the threshold. For a discussion of the choice of such perfectoid algebras, see [Yos25, Section 2.3], and also [BMP⁺24, Lemma 4.23] and [Fay25, Proposition 3.5]

2.2. Ideal containments.

Lemma 2.4. *Let S be an absolutely integrally closed domain, let $z, y, y_1, \dots, y_s \in S$ be elements, let $e \geq 1$ be an integer and $\epsilon \in \mathbf{Q}$ be a rational number.*

(i) *If $\epsilon \in (0, \frac{p}{p-1}]$ then we have*

$$z \in (p^\epsilon, y) \iff z^{1/p^e} \in (p^{\epsilon/p^e}, y^{1/p^e}).$$

(ii) *If $\epsilon \in (0, 1]$, we have*

$$z \in (p^\epsilon, y_1, \dots, y_s) \iff z^{1/p^e} \in (p^{\epsilon/p^e}, y_1^{1/p^e}, \dots, y_s^{1/p^e}).$$

(iii) *If $\epsilon \in (0, \frac{p}{p-1}]$ then we have*

$$z \in (p^\epsilon, y) \implies z^\gamma \in \left(p^{\epsilon\alpha} y^\beta \mid \begin{array}{l} \alpha, \beta \in \frac{1}{p^e} \mathbf{Z} \cap [0, 1) \\ \alpha + \beta = \gamma \end{array} \right)$$

(iv) *If $\epsilon \in (0, 1]$ and $\gamma \in \frac{1}{p^e} \mathbf{Z} \cap (0, 1)$ then we have*

$$z \in (p^\epsilon, y_1, \dots, y_s) \implies z^\gamma \in \left(p^{\epsilon\alpha_0} y_1^{\alpha_1} \dots y_s^{\alpha_s} \mid \begin{array}{l} \alpha_i \in \frac{1}{p^e} \mathbf{Z} \cap [0, 1) \\ \alpha_0 + \dots + \alpha_s = \gamma \end{array} \right)$$

These for $\epsilon \leq 1$ results can be deduced from [Sch12, Proposition 5.5] as the p -adic completion of an absolutely integrally closed domain is perfectoid. We provide a direct and down-to-earth proof for the convenience of the reader.

Proof. We start with statement (i), and we observe that it is enough to prove it in the case $e = 1$. For the ($\implies$) direction, after replacing y with a multiple we may assume that $z = ap^\epsilon + y$ for some $a \in S$, and we look for elements $s \in S$ satisfying $z^{1/p} = sp^{\epsilon/p} + y^{1/p}$ or, in other words,

$$\begin{aligned} ap^\epsilon + y &= (sp^{\epsilon/p} + y^{1/p})^p \\ &= s^p p^\epsilon + p^{1+(\epsilon/p)}(\dots) + y. \end{aligned}$$

The assumption on ϵ ensures that $\epsilon \leq 1 + (\epsilon/p)$, and therefore after canceling the y terms one can divide the above equation by p^ϵ , which ensures that the resulting equation on s is monic. Since S is absolutely integrally closed by assumption, there are elements $s \in S$ satisfying the required equation.

For the ($\impliedby$) direction, after replacing y with a multiple we may assume that $z^{1/p} = sp^{\epsilon/p} + y^{1/p}$ for some $s \in S$, and the equation above gives $z \in (p^\epsilon, y)$.

e As before, it suffices to prove statement (ii) in the case $e = 1$, and we begin with the ($\implies$) direction. After replacing the y_i with multiples, we may assume z takes the form $z = ap^\epsilon + y_1 + \dots + y_s$ for some $a \in S$, and we now look for $s \in S$ satisfying $z^{1/p} = sp^{\epsilon/p} + y_1^{1/p} + \dots + y_s^{1/p}$ or, in other words,

$$\begin{aligned} ap^\epsilon + y_1 + \dots + y_s &= (sp^{\epsilon/p} + y_1^{1/p} + \dots + y_s^{1/p})^p \\ &= s^p p^\epsilon + p(\dots) + y_1 + \dots + y_s, \end{aligned}$$

which once again turns into a monic equation on s after cancellation and division by p^ϵ . The ($\impliedby$) direction also follows from a direct argument, as in part (i).

The proof of statement (iii) is identical to that of (iv), so let us prove (iv). The assumptions on γ mean that we can write $\gamma = c_1 p^{-1} + c_2 p^{-2} + \dots + c_e p^{-e}$ for some integers c_i with $0 \leq c_i \leq p - 1$. Using part (ii), we then obtain

$$z^\gamma = \prod_{i=1}^e (z^{1/p^i})^{c_i} \in \prod_{i=1}^e (p^{\epsilon/p^i}, y_1^{1/p^i}, \dots, y_s^{1/p^i})^{c_i}$$

Note that the i -th factor in the above product of ideals is generated by the monomials

$$p^{\epsilon a_{i0}/p^i} y_1^{a_{i1}/p^i} \dots y_s^{a_{is}/p^i}$$

as the a_{ij} range through all nonnegative integers with $\sum_j a_{ij} = c_i$. We conclude that the product is generated by elements of the form

$$p^{\epsilon \sum a_{0i}/p^i} y_1^{\sum a_{1i}/p^i} y_2^{\sum a_{2i}/p^i} \dots y_s^{\sum a_{is}/p^i},$$

and by setting $\alpha_j = \sum_i a_{ij}/p^i$, we see that all of these are contained in the ideal given. $\square$

Remark 2.5. In statement (iii) and (iv) above, the proof gives an even stronger statement. For example, in (iv), whenever we have $z \in (p^\epsilon, y_1, \dots, y_s)$ we can conclude that z^γ belongs to the ideal generated by the monomials $p^{\epsilon \alpha_0} y_1^{\alpha_1} \dots y_s^{\alpha_s}$ as above, where we impose the extra condition that the α_i add up to γ with no carries in base p .

Before our next remark, we remind the reader that if R is an F -finite regular domain, a pair (R, f^t) is F -regular if there exists $e > 0$ such that

$$R \xrightarrow{1 \mapsto F_*^e f^{\lceil tp^e \rceil}} F_*^e R$$

splits. This is equivalent to requiring that $\tau(R, f^t) = R$. See also [HY03, Tak04, BMS08] noting we do not have an extra test element contribution since R is regular.

Lemma 2.6. *Let k be a perfect field of characteristic $p > 0$, and consider $R := k[[u, x]]$ a formal power series ring in two variables over k . Given integers $a, b \geq 2$, an integer $\ell \geq 1$, and a real number $t > 0$, we have*

$$(R, u^{1-1/a} x^{1-1/b} (u^\ell + x^\ell)^t) \text{ is } F\text{-regular} \iff (R, (u^{a\ell} + x^{b\ell})^t) \text{ is } F\text{-regular}.$$

We give two proofs. First, we point out this is a consequence of general theorems about the behavior of F -singularities under finite maps. For the second, we do a down-to-earth computation.

Proof #1. Consider the extension $A = k[[u', x']] = k[[u^a, x^b]] \subseteq k[[u, x]] = R$ with induced map g on Spec . We know that $(R, u^{1-1/a}x^{1-1/b}(u^\ell + x^\ell)^t)$ is strongly F -regular if and only if

$$(1) \quad \left(A, \frac{a-1}{a} \text{div}(u') + \frac{b-1}{b} \text{div}(x'), (u'^\ell + x'^\ell)^t \right)$$

is. Consider the map $\mathbb{T} : k[[u, x]] \rightarrow k[[u', x']]$ sending 1 to 1 and the other monomial basis elements to zero. The associated ramification-like-divisor is $\mathcal{R}_{\mathbb{T}} := (a-1)\text{div}(u) + (b-1)\text{div}(x)$ as in [ST14]. As

$$g^*\left(\frac{a-1}{a} \text{div}(u') + \frac{b-1}{b} \text{div}(x')\right) - \mathcal{R}_{\mathbb{T}} = 0$$

by [ST14, Theorem 6.25], we see that

$$\mathbb{T}(\tau(R, (u^{a\ell} + x^{b\ell})^t)) = \tau\left(A, \frac{a-1}{a} \text{div}(u') + \frac{b-1}{b} \text{div}(x'), (u'^\ell + x'^\ell)^t\right).$$

Hence, $(R, (u^{a\ell} + x^{b\ell})^t)$ being strongly F -regular implies that so is (1), as $\mathbb{T}$ surjects. The converse follows via an argument of [CR17] as $\mathbb{T}$ sends the maximal ideal of R into the maximal ideal of A . Indeed, if $\tau(R, (u^{a\ell} + x^{b\ell})^t)$ is contained in the maximal ideal of R , then its image is in the maximal ideal of A . $\square$

Proof #2. We first claim that the pair on the left is F -regular if and only if one has

$$(2) \quad (u^\ell + x^\ell)^{\lceil tp^e \rceil} \notin (u^{\lfloor p^e/a \rfloor}, x^{\lfloor p^e/b \rfloor})$$

for some integer $e \geq 0$. Indeed, since we work over a regular ring, the pair is F -regular precisely when the map $R \rightarrow F_*^e R$ given by $F_*^e u^{\lfloor p^e(1-1/a) \rfloor} x^{\lfloor p^e(1-1/b) \rfloor} (u^\ell + x^\ell)^{\lceil tp^e \rceil}$ splits or, equivalently, when

$$u^{\lfloor p^e(1-1/a) \rfloor} x^{\lfloor p^e(1-1/b) \rfloor} (u^\ell + x^\ell)^{\lceil tp^e \rceil} \notin (u^{p^e}, x^{p^e})$$

for some integer $e \geq 0$. The claim then follows from the fact that (u, x) is a regular sequence.

Now let $\tilde{R} := k[[\tilde{u}, \tilde{x}]]$ be a new power series ring, and consider the continuous k -algebra homomorphism $\varphi : R \rightarrow \tilde{R}$ given by $\varphi(u) = \tilde{u}^a$ and $\varphi(x) = \tilde{x}^b$. Note that $\varphi(u^\ell + x^\ell) = \tilde{u}^{a\ell} + \tilde{x}^{b\ell}$ and that for every integer $e \geq 0$ we have

$$(3) \quad (u^{\lfloor p^e/a \rfloor}, x^{\lfloor p^e/b \rfloor}) = \varphi^{-1}((\tilde{u}^{p^e}, \tilde{x}^{p^e})).$$

Therefore, when the pair on the left is F -regular, (2) and (3) give $(u^\ell + x^\ell)^{\lceil tp^e \rceil} \notin \varphi^{-1}(\tilde{u}^{p^e}, \tilde{x}^{p^e})$ for some $e \geq 0$ or, equivalently, $(\tilde{u}^{a\ell} + \tilde{x}^{b\ell})^{\lceil tp^e \rceil} \notin (\tilde{u}^{p^e}, \tilde{x}^{p^e})$, thus giving that the pair on the right is F -regular.

Now assume that the pair on the right is F -regular. By introducing the perturbation $c := \tilde{u}^a \tilde{x}^b = \varphi(ux)$, we know there is some integer $e \geq 0$ such that

$$\varphi(ux \cdot (u^\ell + x^\ell)^{\lceil tp^e \rceil}) = c \cdot (\tilde{u}^{a\ell} + \tilde{x}^{b\ell})^{\lceil tp^e \rceil} \notin (\tilde{u}^{p^e}, \tilde{x}^{p^e})$$

or, equivalently, $ux \cdot (u^\ell + x^\ell)^{\lceil tp^e \rceil} \notin \varphi^{-1}(\tilde{u}^{p^e}, \tilde{x}^{p^e})$. With (3), we get $ux \cdot (u^\ell + x^\ell)^{\lceil tp^e \rceil} \notin (u^{\lfloor p^e/a \rfloor}, x^{\lfloor p^e/b \rfloor})$ and using once again that (u, x) is a regular sequence, we conclude that

$$(u^\ell + x^\ell)^{\lceil tp^e \rceil} \notin (u^{\lfloor p^e/a \rfloor - 1}, x^{\lfloor p^e/b \rfloor - 1}).$$

But this implies condition (2) and so we see that the pair on the left is F -regular by our first claim. $\square$

Remark 2.7. Proof #1 also works in mixed characteristic for a slightly different statement. It shows that $A = \mathbf{Z}_p[[x]]$ and $R = A[p^{\frac{1}{a}}, x^{\frac{1}{b}}]$ then

$$(A, p^{1-1/a}x^{1-1/b}(p^\ell + x^\ell)^t) \text{ is } +\text{- (or BCM-)regular} \iff (R, ((p^{\frac{1}{a}})^{a\ell} + (x^{\frac{1}{b}})^{b\ell})^t) \text{ is } +\text{- (or BCM-)regular}.$$

The point is we have transformation rules for $+$ -test ideals and BCM-test ideals by [MS21].

We now study what a particular blowup of our singularity looks like.

Lemma 2.8. *Suppose $(R, \mathfrak{m} = (u, x), k)$ is a complete regular 2-dimensional local ring and consider $f = u^{a\ell} + x^{b\ell} \in R$ where a and b are relatively prime. Let $\pi : X \rightarrow \text{Spec } R$ be the normalized blowup of the ideal (u^a, x^b) . Set D to be the strict transform of $\text{div}(f)$ on X . Then we have the following.*

- (i) X has a unique reduced exceptional divisor $E \cong \mathbb{P}_k^1$.
- (ii) $-E$ is a π -ample Weil divisor.
- (iii) $K_{X/R} = (a + b - 1)E$ and $\text{div}(f)$ pulls back to $D + (\ell ab)E$.

- (iv) The different of $K_X + E$ restricted to E is $(1 - \frac{1}{a})P_1 + (1 - \frac{1}{b})P_2$ for some points P_1, P_2 on E .
- (v) D intersects E at nonsingular points of X (in particular, away from P_1 and P_2). Explicitly, if we consider the homogeneous coordinate ring $k[y, z]$ for E with $V(y) = P_1$ and $V(z) = P_2$ (0 and ∞), then we have that D restricted to E is given by the equation $(y^\ell + z^\ell)$.

This lemma shows that the normalized blowup of (u^a, x^b) behaves exactly as one expects from the toric setting. Those willing to believe that may wish to skip the proof on first reading.

Proof. Consider the finite extension $S = R[u^{\frac{1}{b}}, x^{\frac{1}{a}}]$ of R (with maximal ideal $\mathfrak{n} \subseteq S$) and consider the map $\varphi : S \rightarrow R$ sending 1 to 1 and the other natural basis elements $u^{i/b}x^{j/a} \mapsto 0$ for $0 \leq i < b$ and $0 \leq j < a$. Consider the maximal ideal $\mathfrak{n} = (u^{1/b}, x^{1/a})S \subseteq S$ and notice that

$$\mathfrak{n}^{ab} \cap R = \varphi(\mathfrak{n}^{ab}) = (u^i x^j \mid i, j \geq 0, ib + ja \geq ab) =: J.$$

As $\mathfrak{n}^{ab}$ is a valuation ideal (associated to blowing up the origin), we see that so is $J \subseteq R$. In particular, J is integrally closed. As R is regular and 2-dimensional, J^n is also integrally closed for all n by a result of Zariski, see [HS06, Theorem 14.4.4]. Let X denote the blowup of J , and Y the blowup of $\mathfrak{n}^{ab}$ (which is the same as the blowup of $\mathfrak{n}$, although the induced relatively very ample divisors have different coefficients), and there is a finite map $\beta : Y \rightarrow X$ induced by the inclusion of Rees algebras $R[Jt] \subseteq S[\mathfrak{n}^{ab}t]$. Note φ extends to these Rees algebras and hence also induces $\varphi : f_*\mathcal{O}_Y \rightarrow \mathcal{O}_X$. It easily follows that Y is smooth, X is normal and Cohen-Macaulay (it is a summand of $f_*\mathcal{O}_Y$), and since Y has a unique reduced exceptional divisor E_Y , we have that X has a unique reduced exceptional divisor E . We have the following diagram.

$$\begin{array}{ccc} E_Y & \xrightarrow{\gamma} & E \\ \downarrow & & \downarrow \\ Y & \xrightarrow{\beta} & X \\ \downarrow \pi_Y & & \downarrow \pi \\ \text{Spec } S & \xrightarrow{\alpha} & \text{Spec } R \end{array}$$

On $K(S) = K(Y)$, the exceptional divisor induces a discrete valuation $v : K(S)^\times \rightarrow \mathbf{Z}$ which takes $u^{\frac{1}{b}} \mapsto 1, x^{\frac{1}{a}} \mapsto 1$, and sends units to 0 (it can also be viewed as the $\mathfrak{n}$ -adic order on S). Consider a monomial $g = u^i x^j = (u^{\frac{1}{b}})^{ib} (x^{\frac{1}{a}})^{ja} \in R$. Notice that $v(g) = ib + ja$. Therefore, as a, b are relatively prime, there exist monomials $h \in K(R)$ with $v(h) = 1$.

We consider the charts of X and Y corresponding to u^a and x^b , which we label X_u, Y_u, X_x , and Y_x respectively. As it is easy to see that $Y_u = \text{Spec } S[x^{\frac{1}{a}}/u^{\frac{1}{b}}] = \text{Spec } S[u^{\frac{i}{b}}x^{\frac{j}{a}} \mid i + j \geq 0, j \geq 0] = \text{Spec } S^u$, one applies φ and sees that $X_u = \text{Spec } R[u^i x^j \mid ib + ja \geq 0, j \geq 0] = \text{Spec } R^u$. Similarly $X_x = \text{Spec } R[u^i x^j \mid ib + ja \geq 0, i \geq 0] = \text{Spec } R^x$.

On Y_u , the exceptional divisor E_Y is defined by the principal ideal $(u^{\frac{1}{b}})$. As $x^{\frac{1}{a}}/u^{\frac{1}{b}}$ is in the ring S^u , $(u^{\frac{1}{b}})S^u$ is also generated as a module over R by the monomials $(u^{\frac{i}{b}}x^{\frac{j}{a}} \mid i + j > 0)S$ (and the analogous result holds for Y_x). Hence, applying φ on X_u (respectively X_x) or using the valuation v , the ideal defining E is generated by the following monomials, even using scalars from R ,

$$I_{E,u} = (u^i x^j \in R^u \mid bi + aj > 0)R \quad (\text{respectively } I_{E,x} = (u^i x^j \in R^x \mid bi + aj > 0)R).$$

A direct computation now easily implies that the exceptional divisor E is isomorphic to $\mathbb{P}_k^1$. Indeed, $x^b/u^a \notin I_{E,u}$, and it is not difficult to see that $R^u/I_{E,u} \cong k[x^b/u^a]$, while $R^x/I_{E,x} \cong k[u^a/x^b]$. By construction, E is an anti-ample Weil divisor. In particular, we see that (i) and (ii) hold.

We observed above that the valuation ring $T \subseteq K(R)$ associated to E (or equivalently to $v|_{K(R)}$) has elements h of value 1. Set $T_Y \subseteq K(S)$ to be the valuation ring associated to v (equivalently, associated to E_Y). Hence $T \subseteq T_Y$ have a common monomial uniformizer h but has an extension of residue fields. Note $\varphi : T_Y \rightarrow T$ sends monomials of T to themselves and is nonzero on the associated residue field extension. We see thus that the divisor associated to $\varphi \in \text{Hom}_T(T', T) = \omega_{T'/T}$ has coefficient zero along E .

We now compute canonical divisors. If we set $K_R = -\operatorname{div}(u) - \operatorname{div}(x)$, then associated to the map $\varphi \in \mathcal{H}\operatorname{om}(S, R)$ we obtain an effective divisor $K_{S/R} = (b-1)\operatorname{div}(u^{\frac{1}{b}}) + (a-1)\operatorname{div}(x^{\frac{1}{a}}) = \frac{b-1}{b}\operatorname{div}(u) + \frac{a-1}{a}\operatorname{div}(x)$ (this is the ramification divisor if p is relatively prime to a and b , otherwise it can be thought of as a replacement for the ramification divisor, see [ST14] for related discussion). We can then set $K_S = \alpha^*K_R + K_{S/R} = -\operatorname{div}(u^{\frac{1}{b}}) - \operatorname{div}(x^{\frac{1}{a}})$. Associated to the blowup π_Y , we see that $K_Y = -E_Y - D_1 - D_2$ where D_1 and D_2 are the strict transforms of $\operatorname{div}(u^{\frac{1}{b}})$ and $\operatorname{div}(x^{\frac{1}{a}})$. Viewing $\varphi \in \mathcal{H}\operatorname{om}(f_*\mathcal{O}_Y, \mathcal{O}_X)$, we then obtain that $K_{Y/X} = (b-1)D_1 + (a-1)D_2 + 0E_Y$ where coefficient of E was computed above. From the formula $K_Y = K_{Y/X} + \beta^*K_X$ we see that $K_X = -E - D_x - D_u$ where D_x and D_y are the strict transforms of $\operatorname{div}(x)$ and $\operatorname{div}(u)$. This is what is expected based on the toric setting.

For (iii), we compute the relative canonical and the pullback of $\operatorname{div}(f)$. Observe that

$$\begin{aligned} \beta^*K_{X/R} &= \beta^*(K_X - \pi^*(-\operatorname{div}(ux))) \\ &= \beta^*(-E - D_x - D_u) + \pi_Y^*(a\operatorname{div}(u^{\frac{1}{a}}) + b\operatorname{div}(x^{\frac{1}{b}})) \\ &= -E_Y - aD_1 - bD_2 + a + bE_Y - aD_1 - bD_2 \\ &= (a+b-1)E_Y \end{aligned}$$

As $\beta^*E = E_Y$, this implies that $K_{X/R} = (a+b-1)E$. Next, we compute the pullback of $\operatorname{div}(f)$. By pulling back to Y and employing a similar argument, we see that $\pi^*\operatorname{div}(f) = (\ell ab)E + D$.

We can now compute the different. Note, $K_X + E$ has coefficient zero along E and so it suffices to compute $(K_X + E)|_E \sim_{\mathbf{Q}} K_E + \operatorname{Diff}$ in a specific way. We do our computation on the chart $U := X_u$ as the other chart is similar. First, we can pick a section $s = x^b/u^a \in \mathcal{O}_U(K_X + E)$. This produces a divisor $D_s = (b-1)D_x|_U \sim (K_X + E)|_U$ on this chart. Consider the image of this section $\bar{s}$ under the canonical map $\omega_U(E) \rightarrow \omega_E$. Viewing these as ideal sheaves, we see that while s does not generate $\omega_U(E) = \mathcal{O}_U(-D_x)$, its image generates the image of ω_E of $\omega_U(E)$, which is the ideal $(x^b/u^a) \in k[x^b/u^a]$. Therefore, we see that the associated divisor $D_{\bar{s}} = 0 \sim K_E$. The different is thus computed as

$$(D_s)|_E = D_{\bar{s}} + \operatorname{Diff}.$$

As $D_s = (b-1)D_x|_U = \frac{b-1}{b}\operatorname{div}(x^b/u^a)$ we see that the different Diff is $\frac{b-1}{b}P_1$ where $P_1 = D_1 \cap E$ (the origin in this chart). The analogous computation shows that on the other chart, the different is $\frac{a-1}{a}P_2$. This proves (iv).

For the final statement (v), we simply need to understand how D , the strict transform of $\operatorname{div}(f)$, restricts to $E \cong \mathbb{P}_k^1$. We again work on the chart $U = X_u$ and observe $f = u^{\ell a} + x^{\ell b} = u^{\ell a}(1 + (\frac{x^b}{u^a})^{\ell})$. As $1 + (\frac{x^b}{u^a})^{\ell}$ does not vanish along the exceptional divisor, we see it must be the equation of the strict transform. As the analogous result holds on the chart X_x , the result follows. $\square$

Proposition 2.9. *Let $R := \mathbf{Z}_p[[x]]$, let $a, b \geq 2$ be integers, and consider the polynomial $f := p^a + x^b \in R$. Consider $R_0 := \mathbf{F}_p[[y, x]]$ and $f_0 := y^a + x^b \in R_0$. Then we have*

$$\operatorname{fpt}(R_0, f_0) \leq \operatorname{ppt}(R, f) \leq \frac{1}{a} + \frac{1}{b}.$$

Proof. We may write $a = \ell A$ and $b = \ell B$, where $\ell = \gcd(a, b)$, so that A and B are relatively prime. Consider the normalized blowup $\pi : X \rightarrow \operatorname{Spec} R$ of (p^A, x^B) as in Lemma 2.8. Note, every $+$ -regular pair is automatically KLT by [MS21, Theorem 6.21] ($\widehat{R}^+$ is big enough for that containment), or use Matlis duality and [BMP⁺23a, Definition 4.16]. Hence, the computation of Lemma 2.8 (iii) implies that

$$\operatorname{ppt}(f) \leq \sup\{t \mid A + B - 1 - t\ell AB > -1\} = \frac{1}{\ell A} + \frac{1}{\ell B} = \frac{1}{a} + \frac{1}{b}$$

as desired. Indeed, it is not hard to see that this is the log canonical threshold.

For the other inequality, let E be the exceptional divisor of π and let D denote the strict transform of $\operatorname{div}(f)$. Suppose now that $t < \operatorname{fpt}(f_0)$. By Lemma 2.6, we see that $t < \operatorname{fpt}(R_0, y^{1-1/A}x^{1-1/B}(u^{\ell} + x^{\ell})^t)$. Using Lemma 2.8 (iv) and (v), we see that $(E, \operatorname{Diff}_E(tD))$ has a section ring whose associated pair is $(R_0, y^{1-1/A}x^{1-1/B}(u^{\ell} + x^{\ell})^t)$. In particular, $(E, (K_X + E + tD)|_E)$ is globally F -regular and so [MST⁺22, Lemma 7.2] implies that (R, f^t) is $+$ -regular (this also follows from [BMP⁺23a, Theorem 7.2]). The result follows. $\square$

Remark 2.10. One can also use direct computation to prove that $\text{ppt}(f) \leq \frac{1}{a} + \frac{1}{b}$. We sketch this here.

Let $\zeta \in R^+$ be a primitive $(2ab)$ -th root of unity, let $e \geq 0$ be an integer and $\gamma \in \frac{1}{p^e}\mathbf{Z}$ be a rational number whose denominator is p^e . We then have

$$f^\gamma = (p^{ab/b} + x^{ab/a})^\gamma = \prod_{i=0}^{ab-1} (p^{1/b} - \zeta^i x^{1/a})^\gamma.$$

Since for every $i = 0, \dots, ab-1$ we have $(p^{1/b} - \zeta^i x^{1/a}) \in (p^{1/b}, x^{1/a})$, by [Lemma 2.4](#), we conclude that

$$f^\gamma \in \left(p^{\beta/b} x^{\alpha/a} \mid \begin{array}{l} \alpha, \beta \in \frac{1}{p^e}\mathbf{Z} \\ \alpha + \beta = \gamma \end{array} \right)^{ab} \subseteq \left(p^{\beta/b} x^{\alpha/a} \mid \begin{array}{l} \alpha, \beta \in \frac{1}{p^e}\mathbf{Z} \\ \alpha + \beta = ab\gamma \end{array} \right).$$

Therefore, whenever $\gamma \geq (1/a) + (1/b)$ we have $ab\gamma \geq a + b$, and hence every choice of α, β with $\alpha + \beta = ab\gamma$ must have $\alpha \geq a$ or $\beta \geq b$, and hence $f^\gamma \in (p, x)R^+$.

Remark 2.11. Suppose $F \in \mathbf{Z}[x, y]$ is homogeneous of degree d , $f := F(p, y) \in \mathbf{Z}_p[[y]] = R$, and $f_0 := F(x, y) \in \mathbf{F}_p[[x, y]]$. Let $X \rightarrow \text{Spec } \mathbf{Z}_p[[y]]$ denote the blowup of the origin with exceptional divisor E . On the chart $\text{Spec } R[y/p]$ corresponding to p , $f = p^d F(1, y/p)$. On $E = V(p)$, the strict transform defined by $F(1, y/p)$ restricts to a divisor defined by $f_0(1, x/y)$. Likewise on the chart corresponding to x , $f = x^d F(1, p/x)$. Restricting the strict transform defined by $F(1, p/x)$ to $E = V(x)$, we get $F(1, y/x)$. Putting this together, we see that the strict transform of $\text{div}(f)$ intersects the exceptional $\mathbb{P}^1$ in $V(f_0)$ (viewed as a homogeneous equation).

The argument of [Proposition 2.9](#) then implies that

$$\text{fpt}(R_0, f_0) \leq \text{ppt}(R, f) \leq \min\{t \mid 1 - td > -1\} = \frac{2}{d}.$$

In fact, the argument is easier as there is no additional contribution to the different to worry about (that is, [Lemma 2.8](#) is unnecessary).

As a consequence, we see that the extremal bounds of [\[KKP⁺22\]](#) also apply in mixed characteristic. Analogous computations can also be done in higher dimension.

3. CUSP-LIKE SINGULARITIES

As usual, $p > 0$ is a fixed prime number, and we let $R := \mathbf{Z}_p[[x]]$ be a power series ring in one variable over $\mathbf{Z}_p$. Given integers $a, b \geq 2$ we let $f := p^a + x^b \in R$, and we let $f_0 := t^a + x^b \in \mathbf{F}_p[[t, x]]$. Recall that we have

$$\text{fpt}(R_0, f_0) \leq \text{ppt}(R, f) \leq \frac{1}{a} + \frac{1}{b}.$$

In particular, we have $\text{ppt}(f) = (1/a) + (1/b)$ whenever $\text{fpt}(f_0) = (1/a) + (1/b)$. We also recall that, by [\[Her14\]](#), whenever $\text{fpt}(f_0) \neq (1/a) + (1/b)$ then the denominator of $\text{fpt}(f_0)$ is a power of p ; that is, there is some integer $e \geq 1$ such that $p^e \text{fpt}(f_0) \in \mathbf{Z}$.

Our general goal in this section is to show that that, when $\text{fpt}(f_0)$ is not too far from $(1/a) + (1/b)$, then we have an equality $\text{fpt}(f_0) = \text{ppt}(f)$. We begin by introducing some notation to set up the technical result. As usual, we assume a prime $p > 0$ is fixed.

Given a real number $\lambda > 0$ and an integer $e \geq 1$, we let $\rho_e(\lambda)$ be the largest rational number whose denominator is p^e with the property that $\rho_e(\lambda) < \lambda$. This can be written out precisely as

$$\rho_e(\lambda) := \frac{\lceil \lambda p^e \rceil - 1}{p^e}.$$

Theorem 3.1. *Let $p > 0$ be a prime number and consider the ring $R := \mathbf{Z}_p[[x]]$. Fix integers $a, b \geq 2$ and consider $f := p^a + x^b \in R$. Fix an integer $e \geq 1$ and a number $\gamma \in (1/p^e)\mathbf{Z} \cap (0, 1)$, and assume that for some integer $c \geq a - \lfloor a/p \rfloor$ one has*

$$\gamma > \frac{1}{c} \left(\rho_e \left(\frac{c}{a} \right) + \rho_e \left(\frac{c}{b} \right) \right).$$

Then $f^\gamma \in (p, x)R^+$.

Proof. Let ζ be a primitive $(2c)$ -th root of unity, and note that

$$f^\gamma = ((p^{a/c})^c + (x^{b/c})^c)^\gamma = \prod_{i=0}^{c-1} (p^{a/c} + \zeta^{2i+1} x^{b/c})^\gamma.$$

For every $i = 0, \dots, c$, we have $p^{a/c} + \zeta^{2i+1} x^{b/c} \in (p^{a/c}, x^{b/c})$ and, by our assumption on c , we know that $a/c \leq p/(p-1)$. We can therefore apply [Lemma 2.4](#) to conclude that

$$(p^{a/c} + \zeta^{2i+1} x^{b/c})^\gamma \in \left(p^{(a/c)\alpha} x^{(b/c)\beta} \mid \begin{array}{l} \alpha, \beta \in \frac{1}{p^e} \mathbf{Z} \cap [0, 1) \\ \alpha + \beta = \gamma \end{array} \right),$$

and therefore f^γ is contained in the c -fold product of these ideals; that is, f^γ belongs to the ideal of R^+ generated by monomials of the form

$$p^{(a/c)(\alpha_1 + \dots + \alpha_c)} x^{(b/c)(\beta_1 + \dots + \beta_c)},$$

where $\alpha_1, \dots, \alpha_c, \beta_1, \dots, \beta_c \in (1/p^e)\mathbf{Z} \cap [0, 1)$ satisfy $\alpha_i + \beta_i = \gamma$ for all $i = 1, \dots, c$. Now, if $f^\gamma \notin (p, x)R^+$, then there is a choice of such α_i and β_i for which $\sum \alpha_i < c/a$ and $\sum \beta_i < c/b$. Since $\sum \alpha_i$ and $\sum \beta_i$ both have denominator p^e , this gives

$$c\gamma = \sum \alpha_i + \sum \beta_i \leq \rho_e\left(\frac{c}{a}\right) + \rho_e\left(\frac{c}{b}\right),$$

contradicting the assumption on γ . $\square$

Corollary 3.2. *Consider the ring $R_0 := \mathbf{F}_p[[y, x]]$ and the element $f_0 := y^a + x^b \in R_0$. Assume $\text{fpt}(R_0, f_0) \neq (1/a) + (1/b)$ and fix an integer $e \geq 1$ such that $p^e \text{fpt}(R_0, f_0) \in \mathbf{Z}$. If there is some integer $c \geq a - \lfloor a/p \rfloor$ for which the inequality*

$$\text{fpt}(R_0, f_0) > \frac{1}{c} \left(\rho_e\left(\frac{c}{a}\right) + \rho_e\left(\frac{c}{b}\right) \right)$$

holds, then $\text{fpt}(R_0, f_0) = \text{ppt}(f)$. In particular, setting $c = a$, we get $\text{fpt}(R_0, f_0) = \text{ppt}(R, f)$ whenever

$$\text{fpt}(R_0, f_0) \geq \frac{1}{a} + \frac{1}{b} - \frac{1}{ap^e}.$$

Proof. Setting $\gamma = \text{fpt}(R_0, f_0)$ in [Theorem 3.1](#), we conclude that $\text{fpt}(R_0, f_0) \geq \text{ppt}(R, f)$. The reverse inequality is given by [Proposition 2.9](#). $\square$

It is shown in [\[Her14\]](#) that, given integers $a, b \geq 2$, the F -pure threshold of $f_0 := t^a + x^b \in \mathbf{F}_p[[t, x]]$ can be computed from the base- p expansions of $1/a$ and $1/b$. We discuss this algorithm in the case where $p > ab$.

For each integer $i \geq 0$, let $x_i \in \{1, \dots, a-1\}$ be the standard representative of p^i in $(\mathbf{Z}/a)^\times$, so that the expansion of $1/a$ in base p is

$$\frac{1}{a} = \frac{1}{p} \left(\frac{px_0 - x_1}{a} \right) + \frac{1}{p^2} \left(\frac{px_1 - x_2}{a} \right) + \frac{1}{p^3} \left(\frac{px_2 - x_3}{a} \right) + \dots$$

Similarly, letting $y_i \in \{1, \dots, b-1\}$ be the standard representative of p^i in $(\mathbf{Z}/b)^\times$, the expansion of $1/b$ in base p becomes

$$\frac{1}{b} = \frac{1}{p} \left(\frac{py_0 - y_1}{b} \right) + \frac{1}{p^2} \left(\frac{py_1 - y_2}{b} \right) + \frac{1}{p^3} \left(\frac{py_2 - y_3}{b} \right) + \dots$$

Now let m be the smallest integer $i \geq 0$ for which $(x_i/a) + (y_i/b) > 1$, with the understanding that $m = \infty$ if this never occurs. Note that, because $x_0 = y_0 = 1$, we always have $m \geq 1$.

This m can also be thought of as follows: the sum $(1/a) + (1/b)$ first carries when performed in base p at the $(m+1)$ -th digit. The F -pure threshold of f_0 is then obtained as a suitable truncation of the sum $(1/a) + (1/b)$ in base p ; more precisely, we have

$$\text{fpt}(f_0) = \frac{1}{p^m} + \sum_{i=1}^m \frac{1}{p^i} \left(\frac{x_{i-1}p - x_i}{a} + \frac{y_{i-1}p - y_i}{b} \right) = \frac{1}{a} + \frac{1}{b} - \frac{1}{p^m} \left(\frac{x_m}{a} + \frac{y_m}{b} - 1 \right),$$

with the understanding that this is $(1/a) + (1/b)$ if $m = \infty$.

Crucially for us, given integers $a, b \geq 2$, there exist rational numbers $C_i \geq 0$ and integers $m_i \geq 1$, indexed by $i \in (\mathbf{Z}/ab)^\times$, such that for all $p > ab$ we have

$$\text{fpt}(f_0) = \frac{1}{a} + \frac{1}{b} - \frac{C_i}{p^{m_i}} \text{ when } p \equiv i \pmod{ab}.$$

By virtue of [Corollary 3.2](#), whenever we have $C_i \leq 1/a$ we can conclude that $\text{ppt}(f) = \text{fpt}(f_0)$ for all primes $p > ab$ for which $p \equiv i \pmod{ab}$. Since the computation of $\text{fpt}(f_0)$ is implemented in the `FrobeniusThresholds` package for `Macaulay2` [\[HSTW21\]](#), we have an effective way to find primes for which we get $\text{ppt}(f) = \text{fpt}(f_0)$; the result of such a search for $2 \leq a, b \leq 5$ are displayed in [Table 1](#) in the Introduction. Note that the full strength of [Corollary 3.2](#) is used for testing some small primes, and that the omission of a given prime from the table does not entail that $\text{ppt}(f)$ differs from $\text{fpt}(f_0)$ —only that we do not know whether it is true.

Remark 3.3. In [Section 4](#) we show that $\text{ppt}(R, f) = \text{fpt}(R_0, f_0)$ for some other cusp-like singularities for which the hypotheses of [Corollary 3.2](#) fail: we show that this is the case for $R := \mathbf{Z}_2[[x]]$ and $f = 2^a + x^2 \in R$ for any $a \geq 2$.

4. SPORADIC EXAMPLES

In this section we explore some sporadic examples, where we make crucial use of the following lemma, which utilizes the fact that the p -adic completion of R^+ is Cohen Macaulay for a power series ring R over the p -adic integers [\[Bha20\]](#).

Lemma 4.1. *Let $R := \mathbf{Z}_p[[x_1, \dots, x_n]]$ be a power series ring over the p -adic integers. Let $\alpha_0, \alpha_1, \dots, \alpha_n \in \mathbf{Q} \cap [0, 1)$, and let $h \in R^+$. Then*

$$p^{\alpha_0} x_1^{\alpha_1} \cdots x_n^{\alpha_n} h \in (p, x_1, \dots, x_n)R^+ \implies h \in (p^{1-\alpha_0}, x_1^{1-\alpha_1}, \dots, x_n^{1-\alpha_n})R^+.$$

Proof. Fix $\gamma_i \in \mathbf{Z}$ the denominator of α_i . Consider $S = R[p^{1/\gamma_0}, x_1^{1/\gamma_1}, \dots, x_n^{1/\gamma_n}]$ also a regular ring. Suppose $p^{\alpha_0} x_1^{\alpha_1} \cdots x_n^{\alpha_n} h \in (p, x_1, \dots, x_n)R^+$. Then certainly $p^{\alpha_0} x_1^{\alpha_1} \cdots x_n^{\alpha_n} h \in (p, x_1, \dots, x_n)\widehat{R}^+$ where $\widehat{R}^+$ is the p -adic completion. As $\widehat{R}^+$ is big Cohen-Macaulay over S by [\[Bha20\]](#), we see that $p^{1/\gamma_0}, x_1^{1/\gamma_1}, \dots, x_n^{1/\gamma_n}$ is a regular sequence on $\widehat{R}^+$ and so

$$h \in (p^{1-\alpha_0}, x_1^{1-\alpha_1}, \dots, x_n^{1-\alpha_n})\widehat{R}^+.$$

Since the ideal on the right contains p , the result follows. $\square$

4.1. The cusp $x^2 + y^3$ in mixed characteristic 2. We consider an honest binomial where p does not replace one of the variables.

Lemma 4.2. *Let $R := \mathbf{Z}_2[[x, y]]$ be a power series ring in two variables over the 2-adic integers. Further suppose that B is a big Cohen-Macaulay R^+ algebra. For $f := x^2 + y^3 \in R$, we have $2^{1/4} f^{1/2} \notin (2, x, y)B$.*

We originally wrote down the proof for $B = R^+$ utilizing [Lemma 4.1](#). However, Linquan Ma pointed out that our argument applies to an arbitrary big Cohen-Macaulay B — a stronger statement yielding [Corollary 4.3](#) below.

Proof. Suppose $2^{1/4} f^{1/2} \in (2, x, y)B$ for a contradiction. Then as B is big Cohen-Macaulay, we have $f^{1/2} \in (2^{3/4}, x, y)B$. Fix a $y^{1/2}$ and $2^{1/4}$ in R^+ and set $g := x + y^{3/2}$. Then we have $f \equiv g^2 \pmod{(2)B}$. This implies that $(f^{1/2} - g)^2 \in (2)B$, and [Lemma 2.4](#) gives that $f^{1/2} - g \in (2^{1/2})B$. On the other hand, we have $f^{1/2} - g \in (2^{3/4}, x, y)$, and by using that B is big Cohen-Macaulay again we conclude that

$$f^{1/2} - g \in (2^{1/2})B \cap (2^{3/4}, x, y)B = (2^{3/4}, 2^{1/2}x, 2^{1/2}y)B.$$

Now write $f^{1/2} = x + y^{3/2} + ax2^{1/2} + by2^{1/2} + c2^{3/4}$. Now, looking at the equation

$$(x + y^{3/2} + ax2^{1/2} + by2^{1/2} + c2^{3/4})^2 = x^2 + y^3$$

modulo $2^{3/2}, x^2, y^2$, we conclude that $2xy^{3/2} \in (2^{3/2}, x^2, y^2)B$, and using that B is big Cohen-Macaulay yet again gives that $1 \in (2^{1/2}, x, y^{1/2})B$, a contradiction. $\square$

Corollary 4.3. *With notation as in Lemma 4.2, we have that $\tau_+(R, f^{1/2+\epsilon}) = R$ for all $1 \gg \epsilon > 0$. In particular, $\text{ppt}(R, f) > 1/2 = \text{fpt}(\mathbf{F}_p[[x, y]], f)$.*

Proof. Fix a sufficiently large perfectoid big Cohen-Macaulay R^+ -algebra B so that we have $\tau_B(R, f^t) = \tau_{\text{BCM}}(R, f^t)$ for all t in the sense of [BMP⁺23b, MS21]. In particular, enlarging B does not change this mixed characteristic test ideal. As R is regular, B is faithfully flat over R (see [HH92, Page 77] or [Bha20, Lemma 2.9]). Thus, arguing as in the proof of Lemma 2.2 (in particular, the implications (vii) $\Rightarrow$ (ii) $\Rightarrow$ (i)), we see that since $f^{1/2} \notin (2, x, y)B$ by Lemma 4.2, that $R \xrightarrow{1 \mapsto f^{1/2}} B$ splits. We thus see that $\tau_{\text{BCM}}(R, f^{1/2}) = R$. It follows from [MS18, Proposition 6.4] that $\tau_{\text{BCM}}(R, f^{1/2+\epsilon}) = R$ for all $1 \gg \epsilon > 0$. But we always have that

$$\tau_{\text{BCM}}(R, f^{1/2+\epsilon}) \subseteq \tau_+(R, f^{1/2+\epsilon})$$

since B is an R^+ -algebra, and so the corollary follows. $\square$

This suggests the following question.

Question 4.4. Is $\text{ppt}(R, f) = 5/6$, or perhaps more interestingly, is it something strictly in between $1/2$ and $5/6$?

4.2. 3 lines. We consider an equation similar to an arrangement of 3 lines, but where p replaces a variable.

Proposition 4.5. *Let $p > 0$ be a prime number and consider the rings $R := \mathbf{Z}_p[[x]]$ and $R_0 := \mathbf{F}_p[[t, x]]$. Let $f = px(p - x)$ and $f_0 = tx(t - x)$. Then we have*

$$\text{ppt}(f) = \text{fpt}(f_0) = \begin{cases} 2/3 & \text{if } p = 3 \\ (2p - 1)/3p & \text{if } p \equiv 2 \pmod{3} \\ 2/3 & \text{if } p \equiv 1 \pmod{3}. \end{cases}$$

Proof. By Proposition 2.9, it suffices to show that $\text{ppt}(f) \leq \text{fpt}(f_0)$, i.e. $f^\alpha \in \mathfrak{m}R^+$ for any $\alpha \geq \text{fpt}(f_0)$. By Lemma 4.1, it suffices to show that

$$(x - p)^\alpha \in (p^{1-\alpha}, x^{1-\alpha}).$$

Suppose $\alpha = a/p^e$. Then by Lemma 2.4, we know that

$$(x - p)^{1/p^e} \in (p^{1/p^e}, x^{1/p^e}).$$

Hence $(x - p)^{a/p^e} \in (p^{1/p^e}, x^{1/p^e})^a$. When $p \equiv 2 \pmod{3}$, we just take $a = (2p - 1)/3$ and $e = 1$. Then by pigeonhole principle, we have $(p^{1/p}, x^{1/p})^{(2p-1)/3} \subset (p^{(p+1)/3p}, x^{(p+1)/3p})$ and hence the desired containment. Otherwise, when $a/p^e \geq 2/3$, we still have $(p^{1/p^e}, x^{1/p^e})^a \subset (p^{(p^e-a)/p^e}, x^{(p^e-a)/p^e})$ by pigeonhole principle.

When $p \not\equiv 2 \pmod{3}$ then it is well known that $\text{fpt}(f_0) = 2/3 = \text{lct}(f)$, and so there is nothing to do. $\square$

4.3. The polynomial $2^a + x^2$ in mixed characteristic 2. Let $R := \mathbf{Z}_2[[x]]$ be a power series ring over the 2-adic integers, let $a \geq 2$ be an integer, and consider the polynomial $f := 2^a + x^2 \in R$. As usual, we consider the equal characteristic 2 analogues $R_0 := \mathbf{F}_2[[y, x]]$ and $f_0 := y^a + x^2 \in R_0$. The polynomial f does not always satisfy the hypotheses of Corollary 3.2 but, nonetheless, we show that $\text{ppt}(R, f) = 1/2 = \text{fpt}(R_0, f_0)$ with a more elementary argument.

Proposition 4.6. *Let $R := \mathbf{Z}_2[[x]]$, $a \geq 2$ be an integer, and $f := 2^a + x^2 \in R$. Then $\text{ppt}(R, f) = 1/2$.*

Proof. From the algorithm in [Her14] discussed above, one sees that $\text{fpt}(\mathbf{F}_2[[y, x]], y^a + x^2) = 1/2$ for every $a \geq 2$. By Proposition 2.9, it is then enough to show that $(2^a + x^2)^{1/2} \in (2, x)R^+$. To do so, it is enough to prove that there is some $s \in R^+$ such that

$$(2^a + x^2)^{1/2} = 2s + x.$$

Squaring both sides and canceling, we see that this equation is satisfied if and only if $s^2 + sx - 2^{a-2} = 0$ which, being monic in s , has a solution in R^+ . $\square$

5. FURTHER QUESTIONS

There is a wide literature of computations of F -pure thresholds which would be natural to explore in mixed characteristic as identified in the introduction. First, we would like to understand the examples $p^a + x^b$ in all mixed characteristics $(0, p)$. Here are the smallest examples for which we do not know the answer:

Question 5.1. What is $\text{ppt}(\mathbf{Z}_3[[x]], x^3 + 27)$? What is $\text{ppt}(\mathbf{Z}_2[[x]], x^4 + 16)$?

Question 5.2. With notation as in this paper, for binomials $f_0 = y^a + x^b \in \mathbf{F}_p[[x, y]]$, and $f = p^a + x^b \in \mathbf{Z}_p[[x]]$, are the only possibilities for the ppt , either $\text{fpt}(f_0)$ or the $\text{lct}(f_0) = \frac{1}{a} + \frac{1}{b}$?

One can ask the analogous questions in higher dimensions (see Lemma 4.2), or for other equations (like line arrangements, see Proposition 4.5). Another interesting target would be elliptic curves and higher dimensional Calabi-Yau hypersurfaces, even diagonal ones such as $x^3 + y^3 + z^3 \in \mathbf{Z}_p[[x, y, z]]$ or $p^3 + y^3 + z^3 \in \mathbf{Z}_p[[y, z]]$, see also [BS15].

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